

Where Did the Aid Go?

AI-Powered Subnational Humanitarian Aid Mapping

Chungmann (Manny) Kim, PhD
Postdoctoral Research Fellow

Department of International Health
Johns Hopkins Bloomberg School of Public Health

AI + Health Care Idea Tank
May 8, 2026

The Unmet Problem

Billions in aid related to humanitarian crises flow annually, but we do not know exactly where they go (Kim 2026, [LINK]).

- The UN Office for the Coordination of Humanitarian Affairs Financial Tracking Service (OCHA FTS) tracks \$10B–\$40B annually in humanitarian funding — but the records do not systematically provide destination location information, and many carry **no sector label** (e.g., health, nutrition, food security).
- Without knowing *where* and *for what* aid was allocated, it **cannot be linked** to regional vulnerabilities, specific intervention needs, or measurable response effectiveness.
- Understanding this gap is especially important as the global humanitarian funding situation is heavily stretched.

The Overlooked Opportunity

The data to build that missing link **already exists** — buried in unstructured text (317,584 records totaling 5,658,681 word tokens as of Feb 2026) — and specialized AI offers a **novel approach** to extract it at scale.

- Humanitarian aid reports contain **implicit geographic references** and **purpose descriptions** in free text — rarely systematically parsed at scale.
- Two AI-solvable gaps: *where did it go?* (geolocating to the subnational level) and *what was it for?* (purpose classification).
- A combined NLP pipeline offers a **scalable, robust solution** to transform these text-based reports into structured, subnationally georeferenced tracking data.

Example: Extracting the Missing Link

Raw Text: “Integrated Nutrition and WASH Drought Response in **Bamyan** Province facing **crisis-level food insecurity**.”
[\$1,000,000]

⇒ **Location:** Bamyan (\$1M) ⇒ **Missing Label:** Food Security, Nutrition, WASH

The AI Mapping Pipeline

A **two-stage AI pipeline** — LLM geoparser + NLP purpose classifier — that converts unstructured upstream humanitarian aid reports into a **geo-located tracking dataset**.

- **Stage 1 — Geolocate:** Context-aware NER models (e.g. **BERT-base**, **GLiNER2**) paired with algorithmic **fuzzy string matching** extract province names; handles transliteration and multi-location fund distributions.
- **Stage 2 — Label & Time:** A trained **Multinomial Naive Bayes** NLP model predicts missing sector keywords based on text context, structuring both the purpose and timing.
- **Stage 3 — Calculate the Gap:** Merge subnational allocations with localized health crisis metrics (e.g., disease outbreak data for Ebola or cholera) to calculate real-time funding vs. need discrepancies.

Stage 1: Georeferencing

“...Response in Bamyan and Daikundi.”
\$1M ⇒ Bamyan \$500k / Daikundi \$500k

Stage 2: Purpose Labeling

“...facing crisis-level food insecurity.”
Missing Label Predicted ⇒ **Food Security**

Precision in Humanitarian Response

This tracking pipeline empowers coordinators and donors to **immediately visualize the gap between emerging subnational needs and actual funding allocations.**

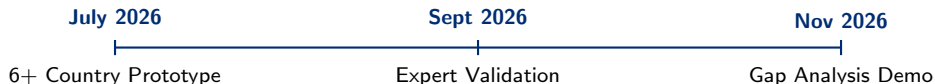
- **Calculate the Gap:** Quantify the **near-real time gap** between arriving funds and subnational crisis needs, directly answering: **does funding align with the disaster?**
- **Crisis Response Targeting:** During acute shocks (e.g., sudden famine warnings), operations can quickly identify which affected districts have secured sector aid vs. which remain completely **unfunded**.
- **Donor Accountability & Redirection:** Disaggregate subnational allocations by donor and purpose to reveal systematic geographic gaps, enabling immediate policy and resource correction.

Next Proof

The 6-Month Milestone

Release an **open, validated subnational humanitarian aid dataset** covering **6+ highly vulnerable countries** — georeferenced, purpose-labeled, and ready for integration.

- **July 2026:** 6+ country prototype — establishing a georeferenced and purpose-labeled subnational dataset ready for regional integration.
- **Sept 2026 (Validation):** Model accuracy actively benchmarked against RA manual ground truth and iteratively vetted through an OCHA expert review loop.
- **Nov 2026 (6-Month Proof):** Dashboard + demonstration of the use case for key selected countries with health needs and gap analysis.



Where Did the Aid Go?

AI-Powered Subnational Humanitarian Aid Mapping

Contact: ckim183@jh.edu

Affiliation: JHU Department of International Health

Key output: Dashboard by Nov 2026 + demonstration of the use case
for key selected countries with health needs and gap analysis

“We spend tens of billions of dollars a year on humanitarian aid — and we have a very limited understanding of localized impact, because we cannot systematically trace it below the country level. AI can fix that.”